

Early Multimodal Physiological Phenotypes and Outcomes in Aneurysmal Subarachnoid Hemorrhage: An Unsupervised Machine Learning Clustering Study Using MIMIC-IV and eICU Databases

Chinese Title: 动脉瘤性蛛网膜下腔出血患者早期多模态生理表型与预后的关联：基于无监督机器学习的表型聚类研究

Study Type: Multicenter retrospective cohort study; unsupervised machine learning-based phenotyping and prognostic risk stratification research

Target Journals: *Critical Care, Neurocritical Care, Critical Care Medicine, Intensive Care Medicine*

1. Background and Rationale

1.1 Clinical Background

Aneurysmal subarachnoid hemorrhage (aSAH) is a critical neurocritical illness characterized by acute onset and poor prognosis, which predominantly affects young and middle-aged adults with high mortality, high disability rates and heavy medical burdens. Within the first 48 hours after ICU admission, patients often develop systemic multimodal physiological disturbances, including neurological deterioration caused by primary intracranial injury, as well as secondary complications such as circulatory instability, tissue hypoxia, early anemia, systemic inflammation and renal dysfunction. The superposition of early multisystem physiological disorders serves as the core risk factor driving short-term adverse outcomes in patients with aSAH.

Early anemia is one of the most common complications in aSAH patients after ICU admission, which is mainly induced by acute stress-induced erythropoiesis suppression, recurrent minor intracranial hemorrhage, surgical trauma, and hemodilution caused by aggressive fluid resuscitation. Multiple previous clinical studies have confirmed that early anemia is closely correlated with elevated mortality risk among aSAH patients. However, prominent contradictions exist in clinical practice: patients with equivalent anemia severity present drastically heterogeneous prognoses. Some anemic patients have stable clinical courses,

whereas others rapidly progress to multiple organ dysfunction syndrome or even death. A single hemoglobin indicator fails to explain such prognostic heterogeneity.

According to preliminary data extraction results of this study, a total of 1,500 aSAH patients were enrolled in the combined dual-database cohort, among whom only 58 patients received red blood cell (RBC) transfusions. The extremely limited sample size of transfusion recipients leads to insurmountable statistical defects for causal inference and subgroup analyses focusing on RBC transfusion strategies, making the original transfusion-oriented research direction statistically unfeasible. Meanwhile, existing studies mostly focus on prognostic analyses of single anemia indicator while ignoring the synergistic effects of multimodal physiological markers including early neurological injury, circulatory status, tissue oxygenation and organ function. The differentiated prognostic value of anemia under varied physiological backgrounds remains unclear, leading to uniform one-size-fits-all anemia management strategies without individualized stratification evidence for aSAH patients.

1.2 Limitations of Previous Studies

Current domestic and international studies regarding early physiological disturbance and anemia prognosis in aSAH have four core limitations. Combined with the real sample distribution of our cohort, the original research direction is completely unworkable:

First, most previous prognostic analyses adopted univariate or limited clinical indicators such as hemoglobin and GCS score separately, which decoupled the inherent correlations among neurological function, circulatory perfusion, tissue oxygenation and renal function. These studies ignored the holistic characteristics and interactive effects of early systemic physiological disorders in aSAH patients, thereby failing to accurately identify high-risk populations.

Second, existing research has not focused on the background-dependent prognostic effects of anemia and failed to answer the key clinical question: the underlying mechanism for divergent outcomes among patients with identical anemia severity. Besides, objective data-driven physiological stratification criteria are lacking, which hinders the formulation of individualized anemia management strategies.

Third, the original transfusion-focused research direction has a fatal sample size defect: only 58 out of 1,500 aSAH patients received RBC transfusions. Such a tiny intervention sample will directly trigger severe machine learning model overfitting, insufficient statistical power and extremely unstable regression results, making analyses on heterogeneous transfusion effects and individualized transfusion subgroups impossible. Therefore, a complete research direction shift is mandatory.

Fourth, traditional clinical stratification relies on empirical manual classification with substantial subjectivity. Data-driven objective stratification based on unsupervised machine learning is scarce, and most existing phenotyping models lack external validation via

multicenter ICU databases, resulting in poor generalizability and limited clinical bedside application value.

1.3 Research Gaps and Core Innovations

Combined with clinical pain points, limitations of previous literature and real sample characteristics of our cohort, four explicit research gaps are identified:

1. No studies have adopted routine bedside multimodal physiological indicators within 48 hours after ICU admission combined with unsupervised machine learning algorithms to objectively explore intrinsic early physiological phenotypes in aSAH patients;
2. The differentiated prognostic impacts of early anemia across distinct early physiological phenotypes remain unclarified, and evidence regarding the background-dependent prognostic effects of anemia is absent;
3. Validated early physiological phenotyping models for aSAH patients with multicenter external verification are lacking. Traditional manual classification is highly subjective with poor repeatability, restricting clinical promotion;
4. Due to the extremely small transfusion sample size, transfusion-based precision medical research is statistically infeasible. A novel prognostic stratification study based on the full cohort with routine physiological indicators is urgently required to maximize the utilization of 1,500 complete patient cases.

Core Innovations of This Study:

1. **Direction innovation: shifting from univariate analysis to multimodal phenotyping:** Breaking the limitation of conventional single-indicator prognostic analysis, this study integrates routine ICU monitoring indicators covering six dimensions (neurology, circulation, oxygenation, anemia, renal function and inflammation) within 48 hours post ICU admission. Unsupervised machine learning is applied to objectively divide patient physiological phenotypes, which fits the actual pathophysiological characteristics of early systemic multisystem disorders in aSAH patients.
2. **Content innovation: revealing background-dependent effects of anemia:** Innovative stratified analyses within each phenotype are conducted to verify that the prognostic value of early anemia varies significantly across different physiological phenotypes. This study explains the core mechanism of prognostic heterogeneity among aSAH patients with equivalent anemia, filling critical research gaps in this field.

3. **Methodological innovation: dual-center external validation combined with interpretable machine learning:** Model training is performed based on the MIMIC-IV database, and external validation is conducted using the independent eICU cohort to guarantee the stability of phenotyping models. SHAP interpretable analysis is further applied to address the black-box problem of machine learning and quantify the contribution of each physiological indicator to phenotype classification and prognostic outcomes.
4. **Dual compatibility with clinical practice and statistical design:** All transfusion-related analyses with small sample defects are abandoned, and the full cohort of 1,500 patients is adopted to completely avoid statistical failure caused by insufficient transfusion cases. All included indicators are routine ICU monitoring parameters with easy data extraction and low missing rates, enabling direct bedside transformation of the final phenotyping model.

1.4 Research Hypotheses

1. Four physiological phenotypes with significantly different prognostic outcomes can be stably identified via unsupervised machine learning based on multimodal routine physiological indicators within 48 hours after ICU admission. The four phenotypes differ distinctly in neurological injury severity, circulatory status, tissue oxygenation, hemoglobin level and short-term clinical outcomes;
2. The prognostic predictive value of early anemia for short-term outcomes including in-hospital mortality and ICU length of stay is significantly phenotype-dependent. Early anemia acts as an independent risk factor in high-risk phenotypes (circulatory failure phenotype and mixed critical phenotype), while it shows no independent prognostic value in the isolated neurological critical phenotype and mild physiological disturbance phenotype;
3. The prognostic prediction efficiency of the multimodal machine learning-derived physiological phenotype combined with anemia status is significantly superior to that of single hemoglobin indicator and traditional clinical scoring systems (GCS, SOFA).

1.5 Research Objectives

1.5.1 Primary Objective

Based on two multicenter critical care databases (MIMIC-IV v2.2 and eICU-CRD v2.0), this study aims to objectively identify early physiological phenotypes of aSAH patients using K-means unsupervised clustering algorithm according to multimodal routine physiological indicators collected within 48 hours after ICU admission. We further compare short-term prognostic differences across distinct phenotypes and explore the heterogeneous prognostic effects of early anemia within each phenotype.

1.5.2 Secondary Objectives

1. To compare the predictive efficiency of machine learning-derived multimodal physiological phenotypes, conventional GCS score and SOFA score for in-hospital mortality in aSAH patients;
2. To quantify the contribution weight of each core physiological indicator to phenotype classification and short-term prognosis via SHAP interpretable analysis;
3. To validate the stability of physiological phenotyping criteria and phenotype-dependent prognostic effects of early anemia in the independent external eICU cohort;
4. To develop a simple bedside physiological phenotyping tool and provide evidence-based references for early individualized monitoring and targeted anemia intervention strategies for aSAH patients.

2. Study Design and Study Population

2.1 Study Design

This is a multicenter, retrospective observational cohort study with dual-cohort external validation, consisting of a training cohort (MIMIC-IV v2.2) and an independent external validation cohort (eICU-CRD v2.0). The whole study complies with the STROBE statement for cohort studies and the TRIPOD statement for prognostic model research. ICU admission time was defined as the unified baseline time point. Extreme values of multimodal physiological indicators within 0–48 hours after admission were extracted for unsupervised machine learning phenotyping. Stratified analyses of anemia-related prognostic effects and dual-cohort validation were performed sequentially to avoid common methodological biases in retrospective studies including immortal time bias and selection bias.

2.2 Data Sources

Fixed database versions for research reproducibility: This study strictly adopts MIMIC-IV v2.2 and eICU-CRD v2.0. All data extraction, cleaning and clustering modeling procedures are matched with the corresponding database table structures without cross-version compatibility. All SQL query scripts, Python data preprocessing codes and machine learning algorithms are archived on GitHub, with complete retention of random seeds, model hyperparameters and missing value processing rules to ensure full research traceability and reproducibility.

Database	Data Description	Research Application
MIMIC-IV v2.2	Single-center ICU data from tertiary hospitals in the United States (2008–2019)	Physiological phenotype clustering modeling, phenotypic characteristic analysis, stratified analysis

		of anemia effects within each phenotype
eICU-CRD v2.0	Multicenter ICU data across the United States (2014–2015)	External validation of phenotyping standards, repeatability verification of interactive effects between anemia and physiological phenotypes

Both databases are publicly available after official review by PhysioNet with complete patient information de-identification. This study belongs to secondary retrospective clinical data analysis, which has been approved by the institutional ethics committee with waived informed consent. All research procedures comply with relevant clinical research specifications.

2.3 Inclusion Criteria

Patients who meet**all** the following criteria will be enrolled:

1. Age ≥ 18 years old; only adult aSAH patients are included, and pediatric patients are excluded;
2. Definite diagnosis of aneurysmal subarachnoid hemorrhage: ICD-9 code (430), ICD-10 codes (I60.0–I60.6), confirmed by surgical records of aneurysm clipping or endovascular embolization; non-aneurysmal subarachnoid hemorrhage cases are excluded;
3. ICU length of stay ≥ 24 hours with complete 48-hour physiological monitoring windows;
4. Integrity of 8 core multimodal physiological indicators $\geq 75\%$ within 48 hours after ICU admission (no more than 2 missing indicators) to guarantee data quality for clustering analysis;
5. Complete baseline demographic and clinical covariate data available at ICU admission.

2.4 Exclusion Criteria

Patients meeting **any one** of the following criteria will be excluded to eliminate confounding factors interfering with phenotyping and prognostic analyses:

1. ICU length of stay < 24 hours without adequate physiological monitoring windows;
2. Massive transfusion patients receiving ≥ 5 units of RBC transfusions within 24 hours after admission, as massive transfusion directly disturbs baseline physiological indicators and objective phenotype classification;
3. ≥ 3 missing core multimodal physiological indicators, failing to meet the requirements of unsupervised clustering modeling;

- 4. Complicated with advanced malignant tumors, end-stage multiple organ failure or confirmed brain death upon admission, whose extremely poor baseline prognosis interferes with research results;
- 5. Cases of traumatic subarachnoid hemorrhage and unexplained non-aneurysmal subarachnoid hemorrhage;
- 6. Patients with multiple ICU admissions; only the first ICU admission record is retained to avoid repeated sample inclusion.

3. Variable Definition, Indicator Extraction and Data Preprocessing

3.1 Core Clustering Indicators (48-hour physiological extreme values after ICU admission, all routinely extractable)

Extreme values of physiological indicators within 48 hours after ICU admission were uniformly extracted, with priority given to raw monitoring data before initial clinical interventions including surgery and blood transfusion to ensure the authenticity of baseline physiological status. A total of 8 core clustering indicators covering six dimensions (anemia, neurological function, circulatory perfusion, tissue oxygenation, renal function, inflammation and coagulation) were included. Detailed indicators, MIMIC-IV item IDs and extraction rules are listed below:

Indicator Dimension	Specific Indicator	MIMIC-IV Item ID	Extraction Rules (0–48h)
Anemia	Minimum hemoglobin (Hb_min)	51222	Minimum value before transfusion within 48h; unified unit: g/dL
Neurological Function	Minimum GCS score (GCS_min)	220739	Minimum neurological score within 48h; data affected by deep sedation were excluded
Circulatory Perfusion	Minimum mean arterial pressure (MAP_min)	220181	Reflecting the lowest level of systemic circulatory perfusion
	Maximum shock index (SI_max)	220045 / 220179	Shock index = heart rate / systolic blood pressure; maximum value within 48h

Tissue Oxygenation	Maximum blood lactate (Lac_max)	50813	Reflecting the severity of systemic tissue hypoxia
Respiratory Oxygenation	Minimum SpO2/FiO2 ratio	220277 / 223835	Non-invasive oxygenation surrogate indicator; minimum value within 48h
Renal Function	Maximum serum creatinine (Cr_max)	50912	Reflecting early renal injury and systemic perfusion status
Inflammation & Coagulation	Minimum platelet count (PLT_min)	51265	Reflecting systemic inflammatory response and coagulation consumption

3.2 Definition of Early Anemia

Referring to critical care and neurocritical care anemia management guidelines, as well as hemoglobin thresholds adopted in the SAHARA randomized controlled trial, early anemia was uniformly defined as: **minimum hemoglobin level < 10 g/dL within 48 hours after ICU admission**. Patients with hemoglobin ≥ 10 g/dL were classified as the non-anemia group. Stratified analyses of the correlation between anemia and prognostic outcomes were conducted within each physiological phenotype subsequently.

3.3 Outcome Definitions

3.3.1 Primary Outcome

All-cause in-hospital mortality: all-cause death occurring at any time during hospitalization. This is a hard endpoint directly extracted from databases without missing data or subjective judgment bias.

3.3.2 Secondary Outcomes

1. All-cause ICU mortality;
2. ICU length of stay, total hospital length of stay;
3. Duration of invasive mechanical ventilation;
4. Incidence of acute kidney injury defined by the 2012 KDIGO criteria;
5. Patient discharge disposition (home discharge, transfer to rehabilitation institution, long-term nursing facility, in-hospital death).

Note: Structured diagnostic data of delayed cerebral ischemia (DCI) and long-term neurological functional outcomes (3-month mRS score) are unavailable in both MIMIC-IV and eICU databases. Therefore, these unreliable endpoints are excluded from all analyses, and only objective and stably extractable outcomes are adopted in this study.

3.4 Unified Data Preprocessing Procedures

1. **Unified time window:** ICU admission was defined as the unified baseline for both databases, and all physiological indicators were limited within the 0–48h window to completely eliminate immortal time bias and temporal confounding;
2. **Unit standardization:** Units of laboratory indicators including hemoglobin and creatinine were unified to eliminate numerical deviations caused by inconsistent unit settings across two databases;
3. **Outlier cleaning:** Abnormal false data caused by equipment input errors were truncated according to clinically acceptable physiological extreme values;
4. **Missing value processing:** Median imputation was adopted for variables with missing rate < 20%, while samples with variable missing rate $\geq 20\%$ were directly excluded. Multiple imputation was avoided to prevent artificial data bias;
5. **Indicator normalization:** All continuous clustering indicators were standardized via Z-score normalization (mean = 0, variance = 1) to eliminate dimensional differences interfering with unsupervised clustering results;
6. **Multiple ICU admission processing:** Only the first ICU admission record was retained for patients with repeated ICU hospitalizations to avoid duplicate sample inclusion.

4. Overall Statistical Analysis and Machine Learning Workflow

The whole analytical procedure consists of four core steps. All data analyses and machine learning modeling were performed using Python 3.9. A two-tailed P value < 0.05 was defined as statistically significant.

4.1 Step 1: Unsupervised Machine Learning for Physiological Phenotyping (Core Analysis)

1. **Clustering algorithm selection:** K-means clustering was adopted as the primary algorithm due to its excellent clinical interpretability and wide application in critical care research; hierarchical clustering was used for supplementary verification of clustering stability;

2. **Optimal cluster number determination:** The elbow method combined with silhouette coefficient was applied to determine the optimal K value. The preset range of cluster numbers was limited to 2–4 categories based on aSAH pathophysiological characteristics;
3. **Phenotype characterization:** Standardized mean values of 8 physiological indicators in each cluster were calculated, and radar charts were plotted to visualize physiological differences across phenotypes. Combined with clinical pathophysiological features, four phenotypes were named clinically: mild physiological disturbance phenotype, isolated neurological critical phenotype, circulatory failure phenotype, and mixed multiple organ critical failure phenotype.

4.2 Step 2: Comparison of Baseline Characteristics and Prognoses Across Distinct Phenotypes

1. Baseline comparison: one-way ANOVA or Kruskal-Wallis H test was used for continuous variables according to data distribution, and Chi-square test was used for categorical variables to compare baseline demographic, clinical and physiological differences across four phenotypes;
2. Prognostic analysis: multivariable logistic regression was performed to adjust for confounding factors including age, gender and baseline disease severity. The mild physiological disturbance phenotype was set as the reference group, and adjusted odds ratios (aORs) for in-hospital mortality were calculated for the other three high-risk phenotypes;
3. Survival analysis: Kaplan-Meier survival curves were plotted for four phenotypes, and the Log-rank test was used to compare inter-group survival differences.

4.3 Step 3: Stratified Analysis of Early Anemia and Interaction Test Within Each Phenotype (Core Innovation)

1. Stratified regression: independent multivariable logistic regression models were constructed within each physiological phenotype. After adjusting for baseline confounders, the independent predictive value of early anemia for in-hospital mortality was analyzed;
2. Interaction test: an interaction term of anemia $\times$ physiological phenotype was introduced to test their interactive effect on clinical outcomes and confirm the background-dependent prognostic effect of anemia;
3. Effect quantification: aOR values corresponding to anemia in different phenotypes were compared to quantify prognostic risk differences of equivalent anemia status under varied physiological backgrounds.

4.4 Step 4: Model Interpretability, External Validation and Sensitivity Analysis

4.4.1 SHAP Interpretable Machine Learning Analysis

The SHAP algorithm was introduced to quantify the contribution weight of 8 physiological indicators to phenotype classification and mortality prognosis, solving the black-box limitation of machine learning and identifying core indicators driving high-risk classification and adverse clinical outcomes.

4.4.2 Multicenter External Validation

Cluster centers generated from the MIMIC-IV training cohort were directly migrated to the independent eICU validation cohort without model retraining for automatic phenotype assignment. The consistency of phenotypic distribution proportion, prognostic trends and anemia interactive effects between two cohorts was verified to evaluate the cross-center generalizability of the phenotyping model.

4.4.3 Multidimensional Sensitivity Analysis

1. Algorithm replacement: hierarchical clustering was used to replace K-means clustering to verify the stability of classification results;
2. Exclusion of transfused patients: all 58 transfusion recipients were excluded to eliminate the interference of RBC transfusion on physiological indicators and clinical outcomes;
3. Time window adjustment: 24-hour physiological extreme values were adopted to replace 48-hour indicators to test the impact of different monitoring windows on research conclusions;
4. Anemia threshold adjustment: dual hemoglobin cutoffs ($Hb < 7$ g/dL and $Hb < 12$ g/dL) were adopted to validate the robustness of anemia-related research findings.

5. Revision Explanation of Research Protocol (Response to Peer Review Comments)

5.1 Fatal Defects of the Original Research Protocol

The original research protocol focused on heterogeneous treatment effects of RBC transfusion strategies based on causal forest machine learning. However, real cohort data demonstrated that only 58 out of 1,500 aSAH patients received RBC transfusions, accounting for less than 4% of the total cohort. Such an extremely small intervention sample inevitably causes severe machine learning model overfitting, biased regression results and completely insufficient statistical power, making the original transfusion-oriented research direction statistically unfeasible.

5.2 Core Revisions of the Current Protocol

1. All original transfusion-related analyses including causal inference, causal forest, CATE individualized treatment effect estimation and transfusion subgroup analyses were completely removed;
2. The core research focus was shifted from heterogeneous transfusion effect analysis to unsupervised clustering of early multimodal physiological phenotypes and phenotype-dependent prognostic effects of early anemia;
3. The full cohort of 1,500 patients was fully utilized without sample waste or small-sample statistical defects;
4. Mature methodological modules including database quality control, data cleaning and temporal standardization were retained to guarantee overall methodological rigor;
5. The peer review response section was retained and updated to elaborate sample size defects and reasons for research direction adjustment.

6. Expected Research Outcomes

1. **Phenotype classification results:** A total of 1,500 aSAH patients can be stably divided into four physiological phenotypes: mild physiological disturbance phenotype (25%), isolated neurological critical phenotype (35%), circulatory failure phenotype (20%), and mixed multiple organ critical failure phenotype (20%). The four phenotypes present significant differences in baseline physiological characteristics;
2. **Inter-phenotype prognostic differences:** In-hospital mortality increases sequentially across the four phenotypes, with an in-hospital mortality rate of approximately 8% in the mild disturbance phenotype and over 45% in the mixed critical failure phenotype, showing statistically significant inter-group prognostic differences;
3. **Core findings of stratified anemia analysis:** Early anemia acts as an independent risk factor for in-hospital mortality only in two high-risk phenotypes (circulatory failure phenotype and mixed critical failure phenotype), while it has no independent prognostic value in the isolated neurological critical phenotype and mild disturbance phenotype. A significant interaction exists between anemia and physiological phenotypes (P for interaction < 0.05);
4. **Model predictive efficiency comparison:** The multimodal machine learning-derived physiological phenotype shows significantly better prognostic predictive performance than single hemoglobin indicator, conventional GCS score and SOFA score;
5. **External validation results:** The eICU external validation cohort can replicate the phenotypic distribution, prognostic differences and anemia interactive effects identified in the MIMIC-IV cohort, verifying favorable cross-center generalizability of the phenotyping model.

7. Strengths and Limitations of This Study

7.1 Study Strengths

1. **Excellent statistical feasibility:** The full cohort of 1,500 patients is adopted for all analyses, completely avoiding statistical failure caused by the tiny transfusion subgroup. Sufficient sample size guarantees stable and reliable results for all statistical tests and machine learning models;
2. **High clinical novelty:** This study is the first to reveal the phenotype-dependent prognostic effects of early anemia in aSAH patients and explain the core mechanism of prognostic heterogeneity among anemic patients, breaking through the limitations of conventional single-indicator analyses;
3. **Strong clinical translational value:** All included indicators are routine bedside ICU monitoring parameters without advanced invasive monitoring equipment. The simple phenotyping criteria can be quickly developed into a bedside scoring tool to guide individualized clinical monitoring;
4. **Rigorous methodological design:** Dual independent cohort external validation combined with multidimensional sensitivity analyses and interpretable machine learning ensures high credibility of research conclusions, meeting the submission standards of top critical care journals;
5. **Optimized classification method:** Empirical manual classification is upgraded to data-driven unsupervised machine learning phenotyping, eliminating subjective bias of traditional clinical stratification.

7.2 Study Limitations

1. Inherent limitations of retrospective observational research: unmeasured confounding factors including unrecorded intracranial pressure fluctuation and individualized bedside treatment strategies cannot be completely eliminated;
2. Subjectivity of unsupervised clustering algorithm: although the optimal K value is determined by combining the elbow method and silhouette coefficient, mild algorithm selection bias cannot be completely avoided;
3. Limitations of public ICU databases: advanced neurological monitoring indicators such as brain tissue partial pressure of oxygen and invasive intracranial pressure are unavailable, lacking direct evidence of cerebral local perfusion;
4. All study data are derived from American ICU populations, so the extrapolation of research conclusions to Asian populations requires further verification via prospective single-center cohorts;

5. Long-term neurological functional outcomes and standardized structured DCI diagnostic data are inaccessible; only short-term survival and hospitalization-related outcomes can be evaluated in this study.

8. Research Timeline

Stage	Core Work Content	Duration
Stage 1	Dual-database data extraction, multimodal indicator cleaning, unit standardization, missing value processing and baseline data quality control	3 weeks
Stage 2	K-means unsupervised clustering modeling, optimal K value screening, phenotype classification and characteristic visualization	2 weeks
Stage 3	Inter-phenotype prognostic comparison, stratified anemia analysis within each phenotype, interaction test and SHAP interpretable analysis	2 weeks
Stage 4	External validation based on eICU cohort, multidimensional sensitivity analysis and comparison of model predictive efficiency	1 week
Stage 5	Complete statistical graph plotting, research result summarization and methodological limitation sorting	2 weeks
Stage 6	Full manuscript writing, revision, code archiving	2 weeks

9. Expert Comment & Study Design Iteration

9.1 Peer Review Comment

Out of 1,500 included aSAH patients, only 58 received RBC transfusions. The extremely small sample size of transfusion recipients leads to severe insufficient statistical power and unstable machine learning model fitting, making the original transfusion-focused research direction statistically infeasible. It is necessary to abandon the transfusion-related core analysis and shift to a novel study direction making full use of the entire cohort.

Single anemia indicator analysis has limited clinical value. It is recommended to conduct multimodal physiological phenotyping based on routine ICU monitoring indicators within 48 hours after ICU admission, explore heterogeneous prognostic effects of early anemia across different physiological phenotypes, and build a more accurate risk stratification system for aSAH patients. All included variables should be routinely extracted from MIMIC and eICU databases without advanced monitoring indicators.

The revised study should adopt unsupervised machine learning for data-driven phenotyping, conduct external validation based on dual databases, and clarify the background-dependent prognostic effect of early anemia, which will improve the clinical novelty and publication potential of this project.

9.2 Response to Reviewer

We fully agree with all comments from the methodologist reviewer. Considering that only 58 patients had RBC transfusion among 1500 overall aSAH cohort, we have completely abandoned the original research focusing on transfusion heterogeneous treatment effect and causal forest model. We redesigned the whole research framework and adopted unsupervised machine learning clustering to identify early multimodal physiological phenotypes of aSAH patients.

The revised study makes full use of all enrolled patients without sample waste, focuses on the background-dependent prognostic value of early anemia, and completes internal modeling and external validation based on two independent ICU databases. All indicators adopted in this study are routine bedside monitoring variables with high data availability, which ensures good clinical conversion value and stable statistical results. The revised protocol has higher clinical innovation and better publication feasibility for critical care journals.

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